

Project Goals and Motivations

The motivation for this project comes from a common challenge in recommender systems: restaurants with very little user engagement are often difficult to recommend effectively. Many recommendation systems depend heavily on past interactions, ratings, and review history. This works well for popular restaurants that already have many reviews, but it creates a disadvantage for newer or less-reviewed restaurants. These restaurants may be relevant to users, but they are less likely to appear in recommendations because the system does not yet have enough interaction data to confidently rank them.

This project focuses on the restaurant recommendation setting using the Yelp Open Dataset. Yelp provides a useful real-world environment because it contains both structured business metadata and user-generated review data. This allows us to explore whether restaurant information such as categories, location, attributes, ratings, and review text can help improve recommendations when interaction history is limited. Instead of relying only on popularity or collaborative filtering signals, our project aims to use content-based and hybrid recommendation methods to better identify restaurants that match a user's preferences.

The main goal of this project is to build and evaluate a cold-start restaurant recommendation system. Specifically, we want to recommend restaurants to users based on their past restaurant preferences while also giving fair consideration to restaurants with few or no prior reviews. The system will use Yelp restaurant metadata and review text to rank candidate restaurants and compare its performance against standard baseline methods such as popularity-based recommendation, collaborative filtering, and matrix factorization. By doing this, we hope to show that cold-start restaurants can still be recommended effectively when useful content signals are available.

Problem Statement

Yelp users face challenges with restaurant discovery during the restaurant's first stage because traditional recommendation systems require data from established dining establishments. The user will find suitable options from several new and niche restaurants which lack existing rating or review data for comparison purposes. The Yelp platform allows users to find restaurants based on their preferences for restaurant type and dining cost and restaurant location and restaurant atmosphere and the quality of restaurant reviews. Cold Start Restaurants are those that have little or no prior user data available for creating training data for the recommenders. The recommender system we will create requires the user's past interaction data and the restaurant's detailed information and the review content to produce the user's Top-N restaurant recommendations. The study will determine whether additional signal processing improves user recommendations compared to the two reference systems which use popularity-based and collaborative filtering techniques.

Justification of novelty using relevant literature

The cold-start recommendation problem continues to present difficulties for recommendation systems because current research shows that collaborative filtering fails to handle new items which lack sufficient user data. The COLING research found that LLMs improve cold-start recommendations but their implementation becomes expensive because organizations need to process extensive product inventories. The research introduces a simple re-ranking system which uses metadata and review information to evaluate restaurants with low customer feedback. The system operates by utilizing comprehensive business details and customer reviews from the Yelp official Open Data Set.

Proposed approach or methodology

Our approach is to construct a top N restaurant recommendation system using Yelp data. The dataset requires filtering to obtain restaurants which we will use to create user-item pairs from customer reviews

and ratings. We implement three initial recommenders: popularity-based recommenders (baseline), matrix factorization recommenders (baseline), and LightFM. The experiment defines businesses which show low customer engagement as cold-start items. The two-stage process generates top-K candidates from the baseline model which we use to re-rank candidate businesses through restaurant metadata and content features from review text to establish a cold-start advantage for businesses. We will assess our models through time-based dataset splits which will help us calculate $\text{recall}@k$ and $\text{ndcg}@k$ and mrr according to our established methods while performing a distinct cold-start assessment.

Dataset(s) and data acquisition methods

Our Main Dataset we plan on using is the Yelp open dataset. This is a publicly available dataset that has real-world data about restaurants. It contains data on business records, user restaurant reviews, ratings, photos, etc. We will access this data directly through Yelps official Open Dataset Release.

The Yelp dataset contains multiple different JSON files, but for this project, we will narrow down our data. We will mainly focus on `business.json` and `review.json`. The business file will give us access to important restaurant-level metadata such as categories, location, attributes, and rating information. The review file has more detailed information about interactions between the user and restaurant, star ratings, timestamps, and review text. If needed, we can plan on adding it to other small fields such as `user.json`, `tip.json`, and `checkin.json`.

For our given recommendation project, we will plan on filtering all the Yelp businesses to just restaurants. Then we plan on also restricting the dataset to a small set of cities to restrict the data. The Yelp data comes in the form of multiple different JSON files. We will load the data and clean it to help it take the form of some kind of structured user interaction table and restaurant feature tables. The goal is to construct a cold-start restaurant recommendation benchmark from the full open data set from Yelp.

Evaluation metrics and plans

For this restaurant ranking task, our metrics will be based more on ranking-based metrics rather than the traditional regression-style statistics we normally use. The primary metrics we plan on using are Hit@K, Recall@K, MRR, and NDCG@K. The Hit@K helps measure if the correct restaurant appears in the top k recommendations. The Recall@K helps measure how often the proper relevant items are able to be retrieved within the top k ranking. MRR helps capture how high up in the ranking the first relevant item is on average. NDCG@K helps measure the overall ranking quality and provides more emphasis/credit to items that are ranked higher on the list.

We will plan on evaluating our methods under both an overall and a cold start configuration. For the overall section we will compare our current proposed model against a set of baseline recommendations on all the different interactions we are able to test. For the cold- start config, we will instead obtain the same metrics on the restaurants, but by using very few actual training interactions. This will help give us insight into how our model performs in general, but also give us insight into how well our model does for restaurants that don't have much history/data.

We plan on using a time-based split for the evaluation protocol, which means that earlier user interactions will be used for training, and then later datapoints will be used for validation/testing. This will help us prevent the problem of leaking from any future interactions and best reflect the real-time use of this kind of recommendation system. We will also compare our proposed method against other standard baseline methods, such as collaborative filtering, matrix factorization recommendation, and popularity-based recommendations. This will help get a good gauge on how our evaluation protocol is behaving and if we need to make any adjustments.